

SSc-MIM

A curated molecular interaction map of diffuse cutaneous systemic sclerosis — construction, patient-data validation, and endotypes

Literature-curated topology · MI2CAST-annotated · SBML L2V4 · validated against four single-cell cohorts

568 species · 308 reactions · 5 modules · 133 SSc-specific curated reactions

Source deck: `docs/presentation.md` → `make presentation` · figures auto-embedded · updated 2026-06-26

Why SSc-MIM

- Systemic sclerosis (SSc) is an autoimmune fibro-inflammatory disease — skin/organ fibrosis, vasculopathy, immune dysregulation — with **no approved disease-modifying therapy**.
- The field lacks a **mechanistic map** tying together the core molecular circuits, which has hampered rational target prioritisation.
- **SSc-MIM** is the first curated, SBGN-compliant Molecular Interaction Map for **diffuse cutaneous SSc**, built in CellDesigner and annotated to MI2CAST.

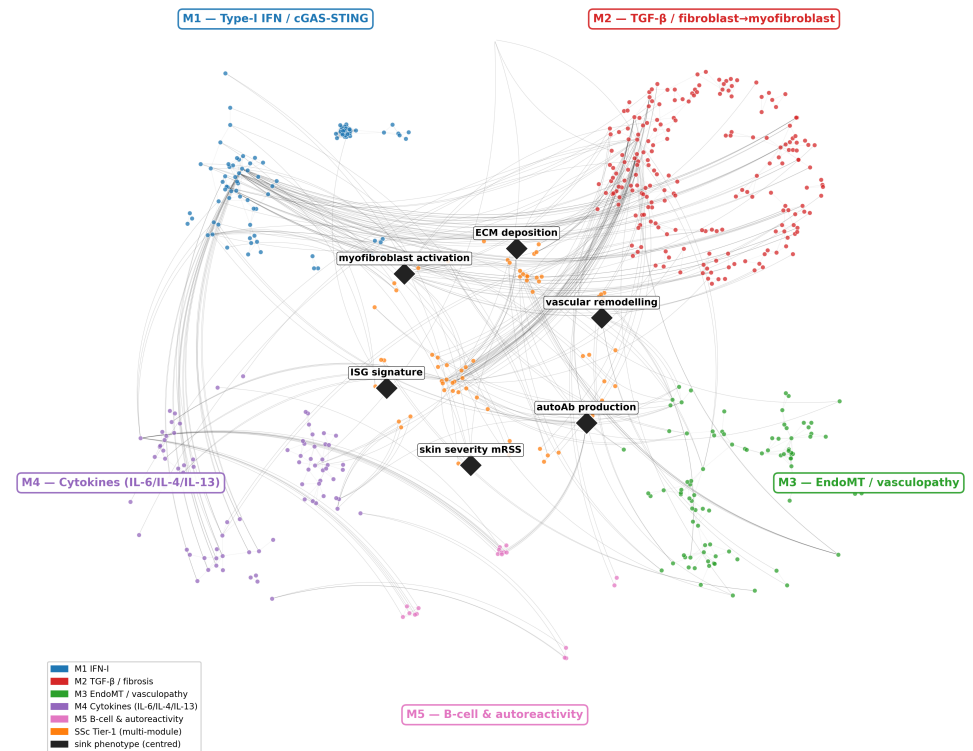
Quantity	Value
Molecular species / reactions / compartments	568 · 308 · 20
Biological modules (+ crosstalk)	5 (M1–M5)
Hand-curated SSc-specific reactions	133 (126 with a primary PMID)
Patient single-cell datasets overlaid	4 — 197 donors (121 SSc / 76 HC)
Map species measurable in patient data	82.6 %
Interactions queued for expert review	143

Map architecture — five modules + crosstalk

- **M1 — Type-I IFN & cGAS–STING.** The innate-immune ignition circuit (ISGs, IRF3/7, STING).
- **M2 — TGF- β / fibroblast→myofibroblast.** The fibrotic core (SMAD2/3/4, collagens, POSTN, COMP).
- **M3 — EndoMT & vasculopathy.** Notch/endothelin, pericyte–endothelial transition.
- **M4 — IL-6 / IL-4 / IL-13 cytokines.** The Th2/JAK-STAT cytokine axis.
- **M5 — B-cell & autoreactivity.** BCR, BAFF/APRIL-BCMA, autoantigens (TOP1, CENPB).
- **Crosstalk layer** wires the modules; **6 phenotype endpoints** (sinks) close the map.

Topology is literature-derived; patient omics are layered on top — they do not define the structure.

Figure 1 — SSc-MIM global view (five-module pentagon layout)
568 species · 1566 edges · modules in fixed positions · phenotype sinks centred · inter-module edges shown as curved arcs



Standards, provenance & honesty

- **Format & tooling:** CellDesigner v4.4, SBGN Process Description, **SBML Level 2 Version 4**; SBML validity enforced by **CI** on every push.
- **Annotation standard: MI2CAST** (identifiers, evidence ECO codes, taxonomy, provenance).
- **Two provenance layers — the honest denominator:**
- *Reactome backbone* (imported, harmonised) vs *SSc-specific curated layer* (the novel content, **133 reactions**).
- **Tiered citation policy (GO-style ECO codes):** 126/133 carry a primary PMID — 73 at direct-assay codes, the rest at expression / physical-interaction / review-traceable codes.
- **Citation integrity:** an AI verdict pass flagged **28 reactions citing an off-topic PMID** (canonical biology, wrong reference); each was replaced with a PMID **verified against live PubMed** — *no citation is ever fabricated*.

The novel content — 133 SSc-specific curated reactions

By module

Module	Reactions
M2 — TGF- β / fibroblast	58
M3 — EndoMT / vasculopathy	27
M1 — Type-I IFN	19
M4 — IL-6/IL-4/IL-13	11
M5 — B-cell / autoreactivity	10
Crosstalk	8

How an interaction is added (gated, never speculative)

- Authoritative PMIDs → fetch OA full text → extract a **verbatim deciding sentence**.
- Candidate edge passes **five anti-nonsense gates G0–G4** (`validate_edge_candidates.py`): identity, grounding, polarity, module-fit, duplication.
- A **negative-control fixture** (fabricated edge) must be rejected by the grounding gate — proving the gate works.
- **Contradiction detection** routes conflicting literature; an **AI verdict pass** gives an advisory validate / revise / caution call.
- **Human ratification is binding** — the curator/co-author decides.

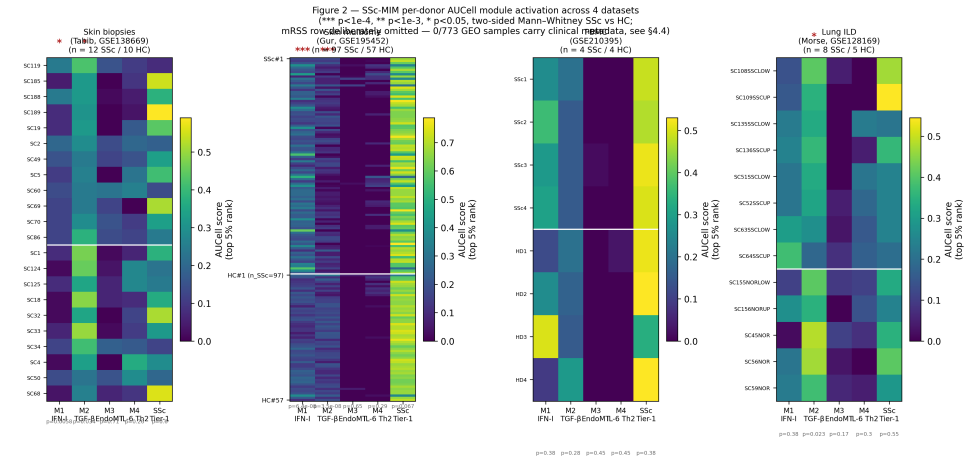
Patient data — four single-cell cohorts

GEO	Reference	Tissue	Donors (SSc/HC)
GSE195452	Gur <i>et al.</i> Cell 2022 (PMID 35381199)	Skin (multiome)	154 (97 / 57)
GSE138669	Tabib <i>et al.</i> Nat Commun 2021 (PMID 34282151)	Skin	22 (12 / 10)
GSE128169	Morse <i>et al.</i> Eur Respir J 2019 (PMID 31221805)	Lung (ILD)	13 (8 / 5)
GSE210395	GEO (no linked publication)	PBMC	8 (4 / 4)

- **Total: 197 donors (121 SSc / 76 HC), 266,884 cells.** Raw archives are **SHA-256-pinned** (`data/MIRROR.md`) for a reproducible input envelope.
- The omics **ground, validate and read out** the map — they **do not define its topology**.

Overlay pipeline & coverage = 82.6 %

- Re-process each dataset from raw counts with **scanpy** (QC, Leiden, cell-type labels — validated vs **CellTypist**, $\kappa = 0.92$).
- **Pseudobulk per donor**; SSc-vs-HC DE with a **mixed-effects negative-binomial GLM** + BH-FDR ($q = 0.05$).
- Map DE genes onto species → **coverage 82.6 %** (195/236): M1 82 · M2 87 · M3 79 · M4 72 · **M5 100** · Tier-1 80 %.
- **Per-donor AUCell** turns each patient into a **5-module activation vector**.

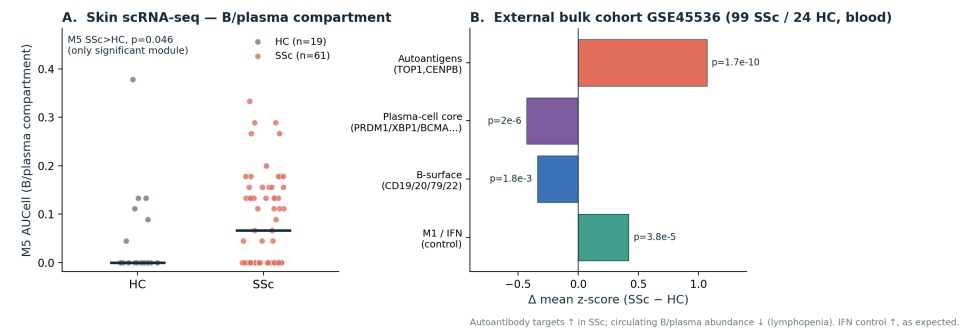


Biological readout — the map behaves like SSc

- **Type-I IFN (M1) is significantly elevated in SSc skin** under sign-blinded AUCell: Gur $p = 6.4 \times 10^{-8}$ ($\Delta = +0.077$), Tabib $p = 5.8 \times 10^{-3}$ — a known SSc hallmark, recovered de novo.
- **M2 (TGF- β)** is active in the fibroblast/myofibroblast compartment across skin *and* lung (FAP, POSTN, CTHRC1, collagens).
- **M3 vasculopathy** surfaces in pericytes (**EDN1** $\uparrow$, **ANGPT2** $\uparrow$), not in whole-tissue endothelium — consistent with EndoMT being a diluted minority state.
- Coverage is reported across a **(significance $\times$ effect-size) grid** (53 % robust $\rightarrow$ 82.6 % permissive), framed as *corroboration*, not validation: a signalling map encodes post-translational events that transcriptomics cannot see.

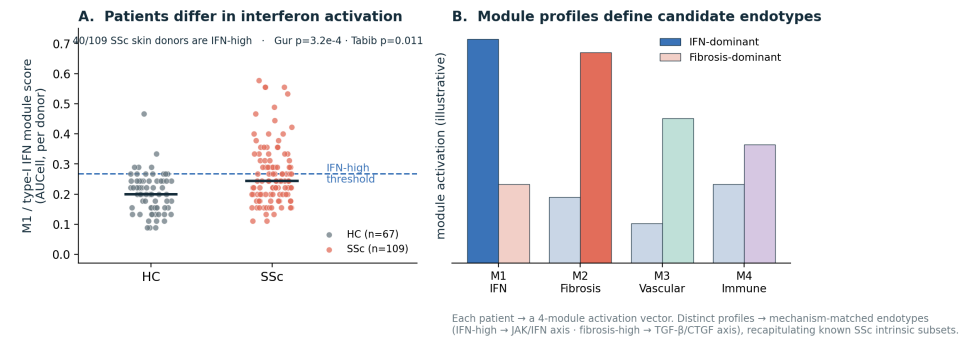
M5 — B-cell & autoreactivity, independently validated

- M5 was split out of the old M4 once its **B-cell / autoantibody** content was disentangled from the IL-6 cytokine core.
- Whole-tissue AUCell for M5 ≈ 0 (B/plasma cells are rare) → validated on a **B/plasma-restricted pseudobulk**:
- **Internal (Gur)**: M5 ScS 0.085 vs HC 0.047, **$p = 0.046$** — the only significant module in that compartment.
- **External (GSE45536, 99 ScS / 24 HC)**: M5 separates ScS/HC **$p = 1.3 \times 10^{-4}$** ; autoantigen core (TOP1/CENPB) **$p = 1.7 \times 10^{-10}$** .



Endotypes — patients as module-activation vectors

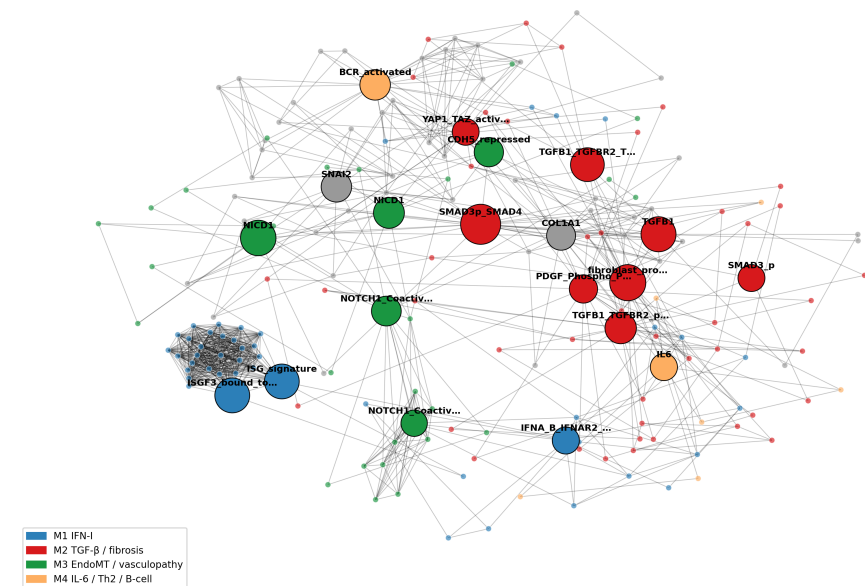
- Each donor's **5-module AUCell vector** is a quantitative, **map-grounded** readout of which circuits are active.
- This exposes candidate **molecular endotypes** (e.g. IFN-high vs fibrosis-dominant) directly interpretable in terms of the curated biology.
- Patient stratification against clinical outcomes (mRSS, autoantibody class) is **reserved as a follow-up**: the public GEO cohorts carry **no per-donor clinical metadata** (the pipeline is built and tested, awaiting an annotated cohort).



Druggable hubs — from topology to targets

- Network analysis (centrality + Leiden communities; **1011 edges, 39 communities**) surfaces hubs that match known SSc biology: **pro-fibrotic state, TGFB1, TGF- β receptor, SMAD3–SMAD4, ISG core**.
- Hubs are cross-referenced against **DGIdb** for druggability and recalibrated against **SSc clinical-trial reality** (e.g. nintedanib, tocilizumab, JAK inhibitors).
- The map is exported as a **Boolean network** (CaSQ $\rightarrow$ SBML-qual) — substrate for a future dynamic-perturbation study.

F3 (preview) — top-20 hub subnetwork in SSc-MIM
size $\propto$ hub_score · colour = source module



Reproducibility & open release

- **Everything is scripted & re-runnable:** `make auto` rebuilds every TSV in `analysis/` and every figure in `figures/` from a clean clone.
- **Container & provenance:** Dockerfile, `environment.yml`, **RO-Crate** metadata, pytest + CI (SBML validation on every push).
- **Inputs pinned:** the four raw scRNA-seq archives are SHA-256-verified (`data/MIRROR.md`).
- **BioModels-ready** SBML (MIRIAM CVTerms injected); **Zenodo DOI** to be minted at the `v1.0` tag.
- **License:** CC-BY-4.0 (map) / MIT (code).

Status & what's next

Done (automated lanes)

- Integrated map, SBML QC, 5-module split (M5 validated).
- Real scRNA-seq overlay on 4 cohorts, coverage 82.6 %, AUCell endotypes.
- Network/druggability, all figures, manuscript (npj-SBA revision v1.1), reviewer swipe-deck app (143 interactions).

Remaining — human-only

- Expert / co-author **adjudication of the 143 interactions**.
- CellDesigner GUI visual round-trip.
- Fill the `.zenodo.json` co-author slot.
- `git tag v1.0` + Zenodo webhook (one-time).

Full written companion: `docs/SSc_MIM_construction_and_validation.md`. *Live status:* `STATUS.md`.